



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each Section (A, B, and C) is to be returned in a separate bundle. Multiple choice questions are to be answered on the answer sheet provided.
- All necessary working should be shown in every question, except multiple choice.

Total Marks – 72

- Attempt questions 1 – 13.
- All questions are not of equal value.
- Unless otherwise directed give your answers in simplest exact form.

Examiner: *A.M.Gainford*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, x > 0$

Multiple Choice

ANSWER ON THE ANSWER SHEET PROVIDED

In Questions 1 to 7 indicate which of the answers A, B, C, or D is the correct answer. Write the letter corresponding to the answer on the answer sheet supplied.

Question 1 (1 mark)

Marks

- The derivative of $\sin^2 x =$
- A:** $2 \sin x$
- B:** $2 \cos x$
- C:** $2 \sin x \cos x$
- D:** $2 \cos^2 x$

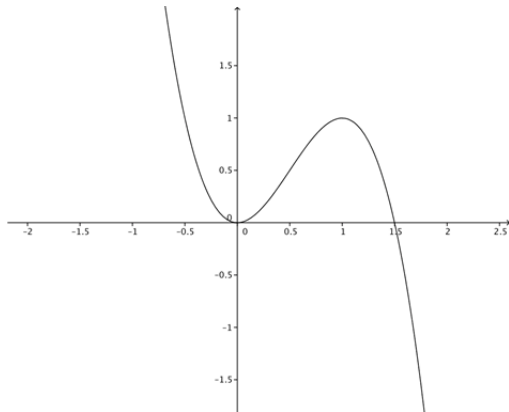
1

Question 2 (1 mark)

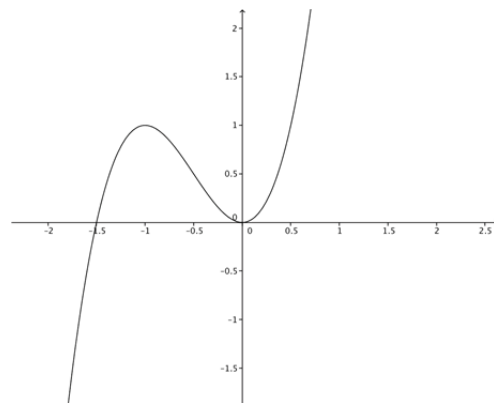
Which of the following best represents the graph of $f(x) = 2x^3 - 3x^2$?

1

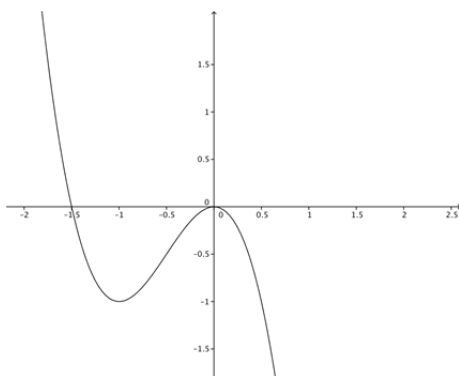
A:



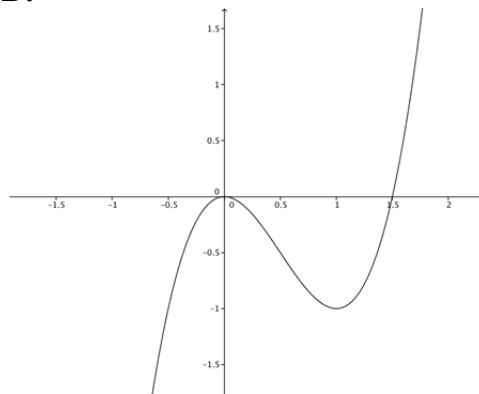
B:



C:



D:



Question 3 (1 mark)

For which values of x is the curve $y = x^3 + 2x^2$ concave up?

1

A: $x < -\frac{2}{3}$

B: $x < -\frac{3}{2}$

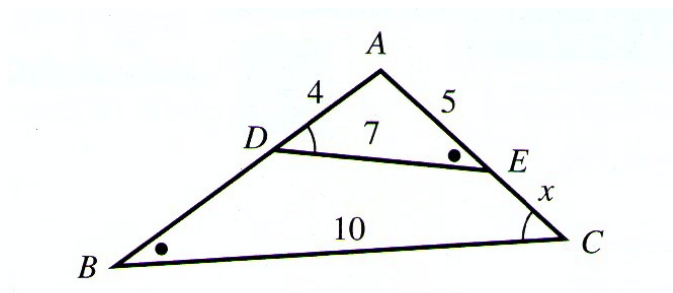
C: $x > -\frac{2}{3}$

D: $x > \frac{3}{2}$

Question 4 (1 mark)

1

In the diagram below, $\angle AED = \angle ABC$ and $\angle ADE = \angle ACB$. The value of x is:



A: $\frac{5}{7}$

B: $3\frac{1}{2}$

C: 5

D: 10

Question 5 (1 mark)

What is the greatest value of the function $y = 6 - 3\cos 2x$?

1

A: 6

B: 3

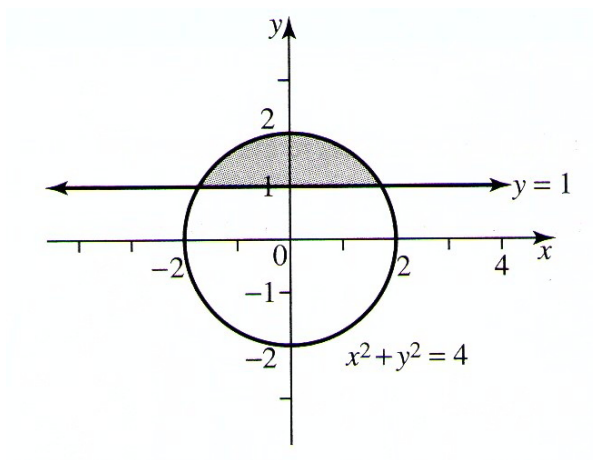
C: 0

D: 9

Question 6 (1 mark)

The shaded region between the circle $x^2 + y^2 = 4$ and the line $y = 1$ is shown below.
The area of the region is given by:

1



A: $\int_{-1}^1 (\sqrt{4-x^2} - 1) dx$

B: $\int_{-1}^1 (\sqrt{4-x^2}) dx$

C: $\int_{-\sqrt{3}}^{\sqrt{3}} (\sqrt{4-x^2}) dx$

D: $\int_{-\sqrt{3}}^{\sqrt{3}} (\sqrt{4-x^2} - 1) dx$

Question 7 (1 mark)

The second derivative of $\frac{e^x + e^{-x}}{2}$ is:-

1

A: $\frac{e^x - e^{-x}}{2}$

B: $\frac{e^{2x} + e^{-2x}}{2}$

C: $\frac{e^x + e^{-x}}{2}$

D: $\frac{e^{2x} - e^{-2x}}{2}$

Section A (20 Marks)
(Start a new booklet)

Question 8 (12 marks)

Marks
5

(a) Differentiate the following:

(i) $y = 3\sin 2x$

(ii) $y = \frac{1}{e^{2x}}$

(iii) $y = x^2 \cos x$

(iv) $y = \frac{\ln x}{x}$

(v) $y = \tan^2 3x$

(b) Find

4

(i) $\int \sin 2x \, dx$

(ii) $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$

(iii) $\int \frac{2x}{x^2 - 7} \, dx$

(c) Use Simpson's rule with five function values to find an approximation to

3

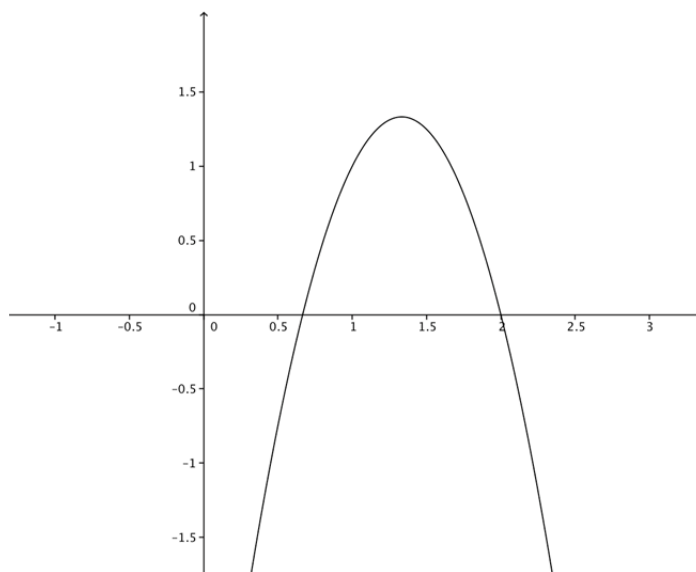
$$\int_0^4 \frac{x^2}{x+1} \, dx$$

(Answer in simplified improper fraction form.)

Question 9 (8 marks)

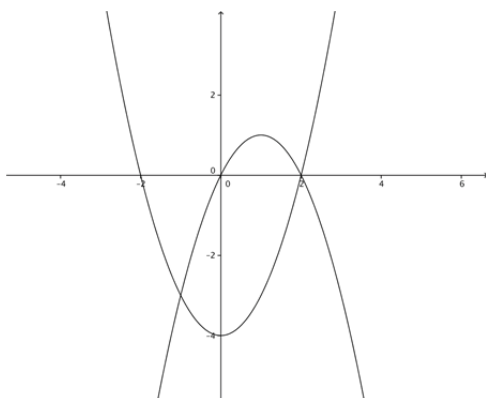
- (a) The graph shows $f'(x)$, the derivative of $y = f(x)$.

4



- (i) Copy the graph to your answer booklet.
- (ii) Sketch on the graph a possible graph of $y = f(x)$ given that the curve passes through the origin. On your graph of $y = f(x)$ mark any points of inflexion.

- (b)



The sketch shows the curves $y = x^2 - 4$ and $y = 1 - (x - 1)^2$.

4

- (i) Find the points of intersection of the curves.
- (ii) Calculate the area of the region between the two curves.

Section B (23 Marks)
START A NEW BOOKLET

Question 10 (11 Marks)

- | | Marks |
|---|--------------|
| (a) Sketch the graph of $y = e^x - 2$ showing the important features. | 2 |
| (b) Find $\int_0^{\frac{\pi}{9}} \sec^2 3x \, dx$ | 2 |
| (c) Sketch the curve $y = 1 + \cos 2x$ in the domain $0 \leq x \leq 2\pi$. | 2 |
| (e) (i) Sketch the graph of $y = 1 + \ln x$, and shade the area bounded by the curve, the y -axis, and the lines $y = 1$ and $y = 2$. | 5 |
| (ii) Make x the subject of the equation $y = 1 + \ln x$. | |
| (iii) Find the volume (in terms of e) generated by rotating the shaded region about the y -axis. | |

Question 11 (12 Marks)

- (a) Consider the function $y = x \log_e x$. 4
- (i) Find the derivative.
 - (ii) Hence find the minimum value of $x \log_e x$ and justify your answer.
- (b) Given the function $y = (3 - x)(x - 2)^2$: 6
- (i) Find the co-ordinates of the stationary points, and determine their nature.
 - (ii) Find the co-ordinates of any points of inflexion.
 - (iii) Sketch the curve in the domain $0 \leq x \leq 4$.
- (c) Find $\frac{d}{dx}(\ln(\cos x))$. Give your answer in simplified form. 2

Section C (22 Marks)

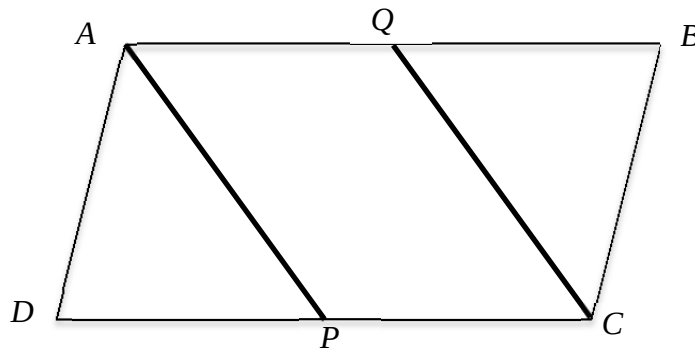
START A NEW BOOKLET

Question 12 (10 Marks)

(a) For the triangle with vertices $P(-1, \frac{1}{2})$, $Q(1, 4)$, and $R(3, 1)$: 5

- (i) Sketch the triangle on a number plane in your answer booklet.
- (ii) Find the midpoint, M , of the interval joining QR .
- (iii) Find the gradient of PM .
- (iv) Show that PM is the perpendicular bisector of QR .

(b) 5



The figure $ABCD$ is a parallelogram. AP bisects $\angle DAB$, and CQ bisects $\angle BCD$.

- (i) Prove that $\triangle DAP \equiv \triangle BCQ$.
- (ii) Prove that $AQ = CP$.

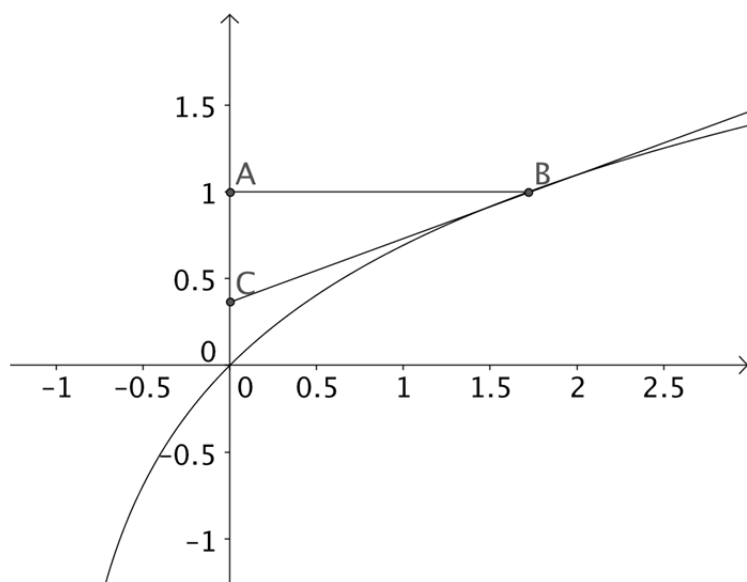
Question 13 (12 Marks)

- (a) Find the equation of the normal to the curve $y = 2 \ln x$ at the point $(e, 2)$. 2

- (b) (i) On the same diagram, sketch the curves $y = \sqrt{x}$ and $y = \frac{1}{\sqrt{x}}$ in the first quadrant. Shade the area bounded by the two curves and the ordinate $x = 2$. 5

- (ii) Find the volume generated when this area is rotated about the x -axis.

- (c) 5



In the diagram, the point B whose y -coordinate is 1, lies on the curve $y = \ln(x+1)$. The tangent to the curve at B cuts the y -axis at C . A straight line through B perpendicular to the y -axis meets the y -axis at A .

- (i) Show that the x -coordinate of B is $(e - 1)$.
- (ii) Show that the equation of the tangent BC is $x - ey + 1 = 0$.
- (iii) Find the length of AC in terms of e .

This is the end of the paper.

Maths 2 unit Task 3

Multiple Choice

- | | |
|---|---|
| 1 | C |
| 2 | D |
| 3 | C |
| 4 | A |
| 5 | D |
| 6 | D |
| 7 | C |

Q8 (a) (i) $y = 3 \sin 2x$
 $y' = 3 \cdot \cos 2x \cdot 2$
 $= 6 \cos 2x$

(ii) $y = \frac{1}{e^{2x}} = e^{-2x}$
 $y' = e^{-2x} \cdot -2$
 $= -2e^{-2x} \quad (= -\frac{2}{e^{2x}})$

(iii) $y = x^2 \cos x$
 $y' = \cos x \cdot 2x + x^2 (-\sin x)$
 $= 2x \cos x - x^2 \sin x$

(iv) $y = \frac{\ln x}{x}$
 $y' = \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2}$
 $= \frac{1 - \ln x}{x^2}$

(v) $y = \tan^2 3x$
 $y' = 2 \tan 3x \cdot \sec^2 3x \cdot 3$
 $= 6 \tan 3x \cdot \sec^2 3x$

(b) (i) $\int \sin 2x \, dx$
 $= -\frac{1}{2} \cos 2x + c$

(ii) $\int_0^{\pi/4} \sec^2 3x \, dx$

PROBLEM: $y = \sec^2 3x$ has a discontinuity at $x = \frac{\pi}{6}$.

Hence it is not possible to calculate $\int_0^{\pi/4} \sec^2 3x \, dx$

If this is not noticed: OR

$\int_0^{\pi/4} \sec^2 3x \, dx$
 $= \frac{1}{3} [\tan 3x]_0^{\pi/4}$
 $= \frac{1}{3} \left\{ \left[\tan \frac{3\pi}{4} \right] - [\tan 0] \right\}$
 $= \frac{1}{3} \left\{ -\frac{1}{\sqrt{2}} - 0 \right\}$
 $= \frac{1}{3\sqrt{2}} - \frac{1}{3}$

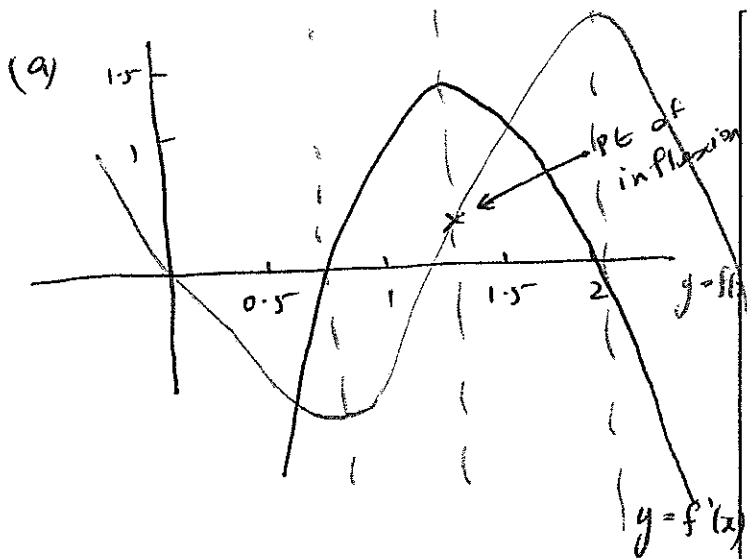
(iii) $\int \frac{2x}{x^2-7} \, dx$
 $= \ln(x^2-7) + c$

(c) $\int_0^4 \frac{x^2}{x+1} \, dx$

x	0	1	2	3	4
y	0	$\frac{1}{2}$	$\frac{4}{3}$	$\frac{9}{4}$	$\frac{16}{5}$

$\approx \frac{1}{3} \left(0 + 4 \times \frac{1}{2} + 2 \times \frac{4}{3} + 4 \times \frac{9}{4} + \frac{16}{5} \right)$
 $= \frac{1}{3} \left(2 + \frac{8}{3} + 9 + \frac{16}{5} \right)$
 $= \frac{253}{45}$

Q 9



2 st pts

1 pt of inflection

1 shape + origin

(b) (i) $y = x^2 - 4$ $y = 1 - (x-1)^2$

For pt of $\cap$: $x^2 - 4 = 1 - (x-1)^2$

$$x^2 - 4 = 1 - x^2 + 2x - 1$$

$$x^2 - 4 = -x^2 + 2x$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0 \quad 2$$

$$\therefore x = 2 \text{ or } x = -1$$

$\therefore$ pts of $\cap$ are $(-1, -3)$ and $(2, 0)$

(ii) Area = $\int_{-1}^2 \left(1 - (x-1)^2 - (x^2 - 4) \right) dx$

$$= \int_{-1}^2 (1 - x^2 + 2x - 1 - x^2 + 4) dx$$

$$= \int_{-1}^2 (4 + 2x - 2x^2) dx$$

$$= \left[4x + x^2 - \frac{2}{3}x^3 \right]_{-1}^2 \quad 2$$

$$= \left[8 + 4 - \frac{16}{3} \right] - \left[-4 + 1 + \frac{2}{3} \right]$$

$$= 15 - 6$$

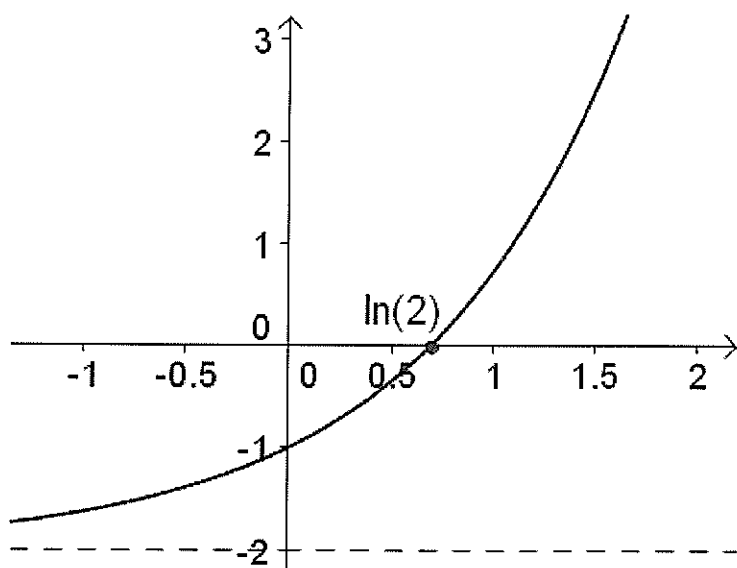
$$= 9 \text{ units}^2$$

2013 Maths 2 unit Task 3

Section B

Question 10

(a)



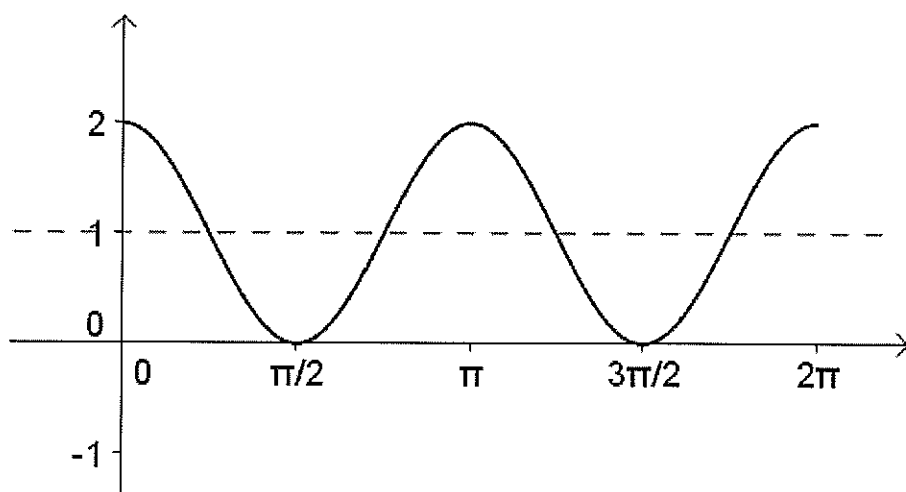
(b)

$$\int_0^{\frac{\pi}{9}} \sec^2 3x \, dx = \frac{1}{3} [\tan 3x]_0^{\frac{\pi}{9}}$$

$$= \frac{1}{3} \left(\tan \left(\frac{\pi}{3} \right) - \tan 0 \right)$$

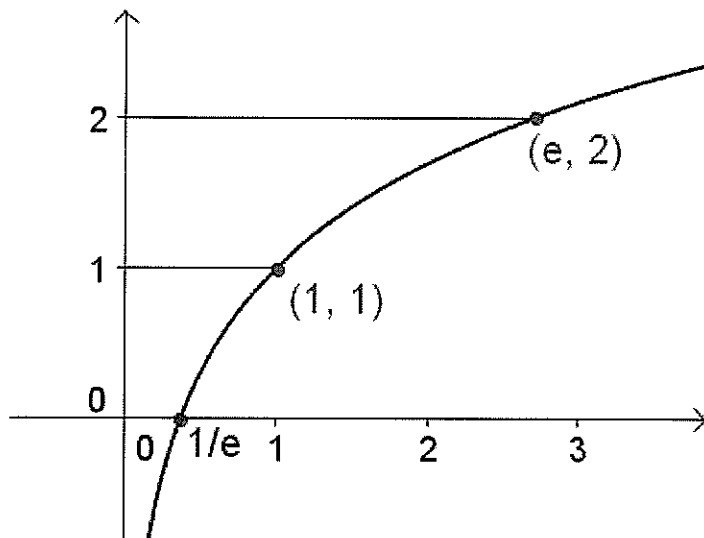
$$= \frac{\sqrt{3}}{3}$$

(c)



(e)

(i)



(ii)

$$y = 1 + \ln x$$

$$y - 1 = \ln x$$

$$e^{y-1} = x$$

(iii)

$$\pi \int_1^2 (e^{y-1})^2 dy = \pi \int_1^2 e^{2y-2} dy$$

$$= \frac{\pi}{2} [e^{2y-2}]_1^2$$

$$= \frac{\pi}{2} (e^2 - 1) \text{ units}^2$$

Q11

(a) (i) $y = x \log_e x$

$$y' = \log_e x + 1$$

(ii) $y'' = \frac{1}{x}$

Stat pts ($y' = 0$)

$$y \log_e x + 1 = 0$$

$$\log_e x = -1$$

$$x = \frac{1}{e}$$

$$y'' = e \text{ at } x = \frac{1}{e}$$

$$y'' > 0 \quad \forall x \quad \therefore \text{minima}$$

minimum occurs at $x = \frac{1}{e}$

$$y = \frac{1}{e} \log_e \frac{1}{e}$$
$$= -\frac{1}{e}$$

(b) (i) $y = (3-x)(x-2)^2$

$$y' = -(x-2)^2 + 2(3-x)(x-2)$$

$$= -x^2 + 4x - 4 + 2(3x - 6 - x^2 + 2x)$$

$$= -3x^2 + 14x - 16$$

Start Pts ($y'=0$)

$$\frac{(3x-6)(3x-8)}{3} = 0$$

$$(x-2)(3x-8) = 0$$

$$x = 2, \frac{8}{3}$$

$$y' = -6x + 14$$

Nature at $x=2$

$$y'' = -12 + 14 = 2 > 0 \text{ minimum.}$$

Nature at $x = \frac{8}{3}$

$$y'' = -16 + 14 = -2 < 0 \text{ maximum}$$

At $x=2$, $y=0$

At $x = \frac{8}{3}$, $y = \frac{4}{27}$

$(2, 0)$ minimum

$(\frac{8}{3}, \frac{4}{27})$ maximum.

$$(ii) \quad y'' = -6x + 14$$

$$-6x + 14 = 0$$

$$x = \frac{7}{3}$$

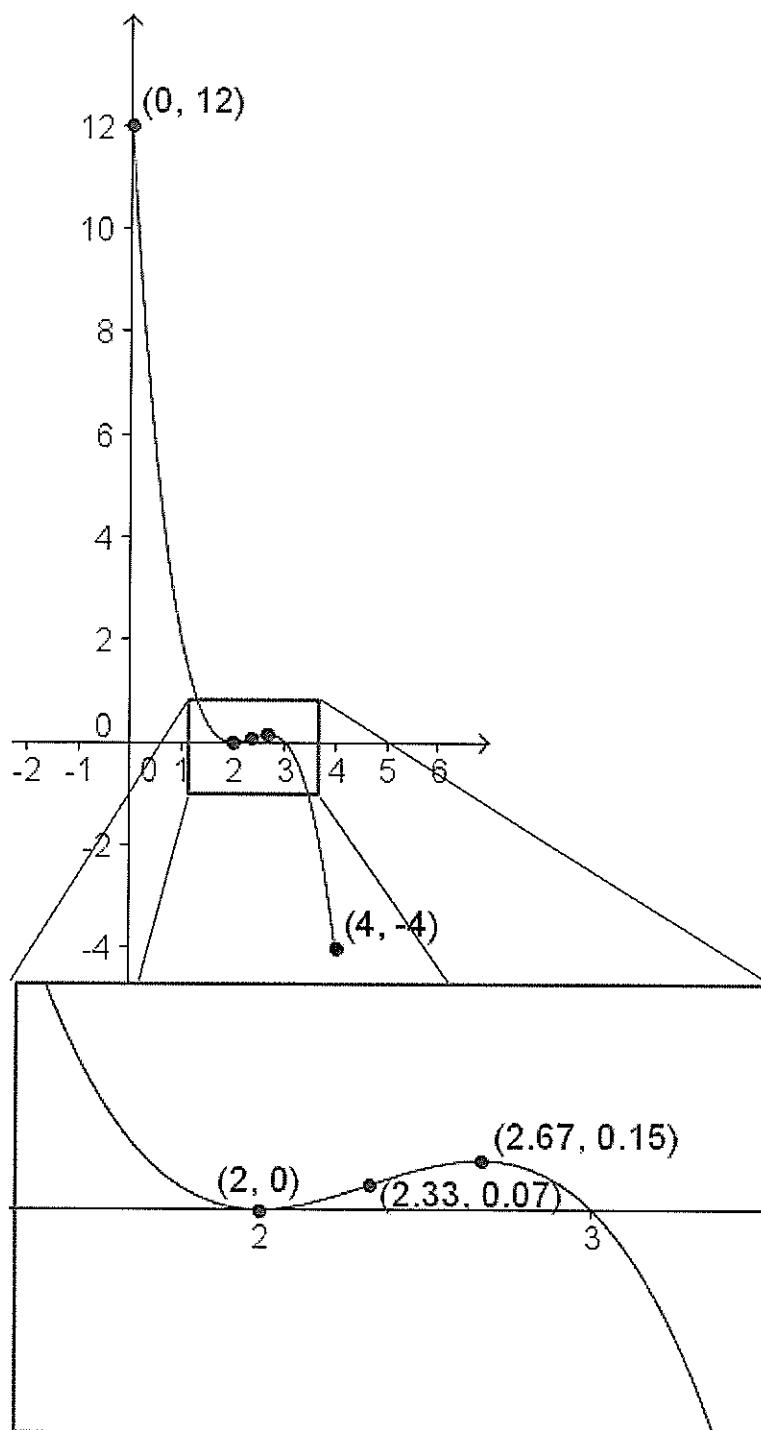
At $x = \frac{7}{3}$, $y = \frac{2}{27}$

Inflexion $(\frac{7}{3}, \frac{2}{27})$

(iii)

At $x = 0, y = 12$

At $x = 4, y = -4$



(c)

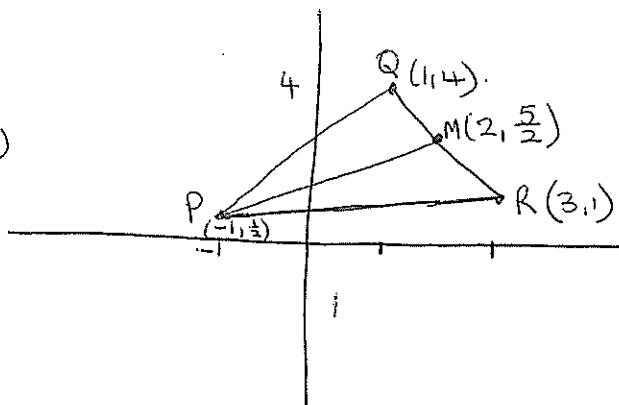
$$\frac{d}{dx} \ln(\cos x) = -\frac{\sin x}{\cos x} = -\tan x$$

2013 2U TASK 3U Q12+13.

12

a)

(i)



①

(ii)

$$M_{QR} = \left(\frac{1+3}{2}, \frac{4+1}{2} \right) = \left(2, \frac{5}{2} \right) \quad \text{①}$$

$$(iii) \quad m_{PM} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{5}{2} - \frac{1}{2}}{2 - (-1)} = \frac{2}{3} \quad \text{①}$$

$$(iv) \quad M\left(2, \frac{5}{2}\right) \text{ bisects } QR. \quad \triangle \quad \text{②}$$

$$\text{Gradient} = m_{QR} = \frac{4-1}{1-3} = -\frac{3}{2}.$$

$$m_{PM} = -\frac{1}{m_{QR}} \quad \triangle \quad \text{perpendicular bisector.}$$

b) In $\triangle DAP$ & $\triangle BCQ$

A: $\angle DAP = \angle BCQ$ (opp $\angle$'s of $\parallel$ gram are equal then bisected).

S: $AD = BC$ (opp sides of $\parallel$ gram are equal).

A: $\angle APD = \angle QBC$ (opp $\angle$'s of $\parallel$ gram are equal).

$\triangle DAP \equiv \triangle BCQ$.
AAS. ③

By corresp. sides of congruent $\triangle$'s.

$$QB = DP.$$

$AB = DC$ (opp sides of $\parallel$ gram equal)

$$\therefore AB - QB = DC - DP. \quad \text{②}$$

$$AC = CP \quad \text{QED.}$$

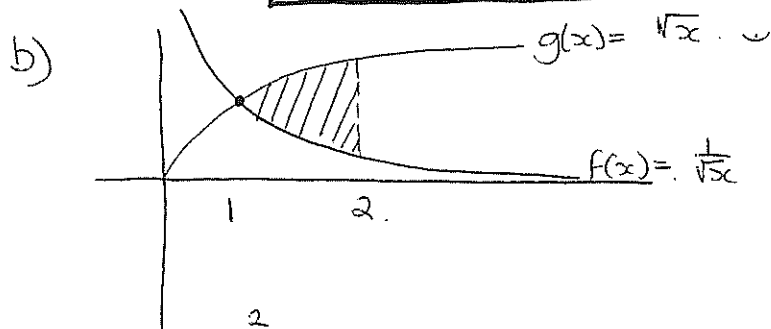
13 a) $y = 2 \ln x$ $\frac{dy}{dx} = \frac{2}{x}$ at $(e, 2)$.

Perpendicular = $-\frac{dx}{dy} = -\frac{x}{2}$.

$y - y_1 = m(x - x_1)$

$y - 2 = -\frac{e}{2}(x - e)$

$y = \frac{e^2}{2} - \frac{ex}{2} + 2$



$V = \pi \int_1^2 g(x)^2 - f(x)^2 \cdot dx$

$= \pi \int_1^2 x - \frac{1}{x} \cdot dx$

$= \pi \left[\frac{x^2}{2} - \ln x \right]_1^2$

$= \pi \left[\left(\frac{4}{2} - \ln 2 \right) - \left(\frac{1}{2} - \ln 1 \right) \right]$

$= \pi \left[1\frac{1}{2} - \ln 2 + \ln 1 \right] = \pi \left[\frac{3}{2} - \ln 2 \right] \text{ units}^3$

13 (c) $y = \ln(x+1)$

$1 = \ln(x+1)$

$e^1 = e^{\ln(x+1)}$

$e = x+1$

$x = e - 1$

$\frac{dy}{dx} = \frac{1}{x+1} = \frac{1}{e} = m$

$(e-1, 1)$

$y - y_1 = m(x - x_1)$

$y - 1 = \frac{1}{e}(x - (e-1))$

$ey - e = x + 1 - e$

BC is $x + 1 - ey = 0$ ✓

BC cuts x-axis when $x=0$.

$ey = 1$

$y = \frac{1}{e}$

Co-ords A(0, 1) C(0, $\frac{1}{e}$)

Distance $AC = 1 - \frac{1}{e}$ units